

Akshat Sharma

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EDUCATION

Thapar Institute of Engineering and Technology, Patiala.
Bachelor Of Engineering in Computer Science.

Sep 2022 – Aug 2024
CGPA: 8.84/10.00

City Montessori School, Lucknow.
Higher Secondary Certificate.

Apr 2019 – Apr 2021
GPA: 99.25/100

PROJECTS

- AI based Rate Limiter | Nodejs, Redis, Worker Threads, Flutter, ML, K6 | [GitHub](#)** *May 2025-Jul 2025*
- Built a concurrent high-performance rate limiter with Redis and worker threads for managing burst API loads.
 - Designed anomaly detection using Z-score for adaptive throttling and auto-banning of malicious IPs.
 - Simulated 100 virtual users across five traffic profiles using k6 to generate **5000 requests in 10s, achieving 84% blocking efficiency** through dynamic throttling and anomaly detection.
- Pronunciation Checker | FastAPI, Transformers, PyTorch, TorchAudio | [GitHub](#)** *Apr 2025-May2025*
- Built a web-based pronunciation system with IPA feedback at phoneme granularity using self-supervised ASR.
 - Utilized Wav2Vec2.0 and Branchformer (Conv-Transformer) for high-resolution speech embeddings.
 - Optimized scoring pipeline to reach **8% WER and 91% ROC AUC** on phoneme classification benchmarks.
- Price Pulse | Flutter, Dart | [GitHub](#)** *Feb 2024-Apr 2024*
- Developed a price-comparison portal that scrapes commodity prices from multiple e-commerce websites.
 - Scraped web pages from scratch using only html parser and http packages.
 - Implemented a secure authentication system using JWTs

EXPERIENCE

- Flutter Developer Intern | Medlr** *Jun 2025-Jul 2025*
- Developed a scalable event and community interaction platform including an admin panel.
 - Implemented video streaming using S3 and CloudFront, improving **loading speeds by 40%**.
 - Built a responsive, cross-platform UI with real-time updates and interactive community forums.
- ML Intern | Thapar Satellite Laboratory** *Jun 2024-Jul 2024*
- Built a time series model to predict cyclone formation in the Indian Ocean with **81% accuracy**.
 - Engineered a comprehensive dataset by extracting four key meteorological parameters from APIs.
 - Leveraged insights from scientific papers to select optimal parameters influencing cyclone development.

SKILLS

Domains: Full Stack Development, App Development, Software Development.

Languages: Java, Dart, JavaScript, Python, C, C++, SQL, HTML.

Frameworks: Flutter, Node.js, Angular, Express, FastAPI, TensorFlow, PyTorch, Grafana K6, Docker.

Databases: MySQL, MongoDB, OracleDB Redis.

EXTRA CURRICULAR ACTIVITIES

Secondary High School Certificate: Secured 2nd All India Rank in 12th grade board exams with 99.25%.

High School Basketball: Runner-up at Inter-House Championship.